

Earned Value Management Cheat Sheet

The formulas, how to read them and the checks to run before you trust them. Full explanation and a worked example: [Earned value management explained](#).

The core values

Term	Meaning
PV planned value (BCWS)	Budgeted cost of the work <i>scheduled</i> to be done by the data date.
EV earned value (BCWP)	Budgeted cost of the work <i>actually completed</i> by the data date, measured by agreed rules of credit.
AC actual cost (ACWP)	Cost actually incurred for the completed work, including accruals for work done but not yet invoiced.
BAC budget at completion	Total approved budget for the work (the end point of the PV curve).
EAC estimate at completion	Expected total cost when the work is finished.

Variances and indices

Measure	Formula	Reading
Cost variance (CV)	$EV - AC$	Negative = over cost for the work achieved
Schedule variance (SV)	$EV - PV$	Negative = behind plan (in value terms)
Cost performance index (CPI)	$EV \div AC$	Below 1.0 = each unit spent earns less than planned
Schedule performance index (SPI)	$EV \div PV$	Below 1.0 = progressing slower than planned
Percent complete	$EV \div BAC$	Achievement — not the same as percent spent
Percent spent	$AC \div BAC$	Spend — compare with percent complete

Forecasting

Method	Formula	Use when
EAC — CPI method	$BAC \div CPI$	Current cost efficiency is expected to continue
EAC — CPI × SPI method	$AC + (BAC - EV) \div (CPI \times SPI)$	Schedule pressure will also drive cost
EAC — remaining at budget	$AC + (BAC - EV)$	The cause of past variance will not recur
EAC — bottom-up	AC + new estimate to complete	Original assumptions no longer hold
Variance at completion (VAC)	$BAC - EAC$	Negative = forecast overrun
To-complete performance index (TCPI)	$(BAC - EV) \div (BAC - AC)$	Efficiency needed on remaining work to finish on budget

Present two or three methods as a range and say which one you recommend and why. More detail: [Estimate at completion formulas](#).

Rules of credit (progress measurement)

- **Units complete** — counted quantities (metres, drawings, units).
- **Percent complete** — estimated; pair with evidence and caps.

- **Weighted milestones** — agreed steps, e.g. received 10%, installed 40%, connected 25%, tested 15%, commissioned 10%.
- **0/100, 50/50, 25/75** — short activities only (one or two periods).
- **Level of effort** — support work; always SPI 1.0, so keep it small.
- **Apportioned effort** — tied to the work it supports (e.g. inspection).

Agree the method before work starts and phase the baseline the same way. Guide: [Progress measurement methods](#).

Worked mini-example

BAC 2,000,000 · PV 1,000,000 · EV 930,000 · AC 1,050,000
 CV = -120,000 · SV = -70,000 · CPI = 0.886 · SPI = 0.930
 EAC (CPI) ≈ 2,257,000 · EAC (CPI × SPI) ≈ 2,348,000 · TCPI ≈ 1.126

Reading: cost efficiency is about 89%; to finish on budget the remaining work must run at a CPI of about 1.13 — usually a sign a formal re-forecast is needed.

Five-question health check — before you present CPI or SPI

- ☐ Were rules of credit agreed before work started, and is progress evidenced?
- ☐ Does AC include accruals for work done but not yet invoiced?
- ☐ Is the baseline current and approved, with changes only through change control?
- ☐ Is level-of-effort work separated from discrete work?
- ☐ Late in the project, is SPI supplemented by earned schedule or critical-path analysis?

Common traps

- Confusing percent spent with percent complete.
- Building PV from a budget spread evenly instead of from the schedule.
- Re-baselining to remove variances caused by poor performance.
- Reporting one month's index without the trend.

Related reading: [The S-curve in project management](#) · [How to build a cost baseline](#) · [Earned schedule explained](#).

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